

DATA VISUALIZATION — CASE STUDY REPORT

Power BI Stock Market Dashboard: Price & Volume Trend Analysis

*A visual walkthrough of an interactive Power BI report tracking
stock price movement, trading volume and moving averages*

NAME : D.MONICA INFANTA
ROLL NO : E25DA028

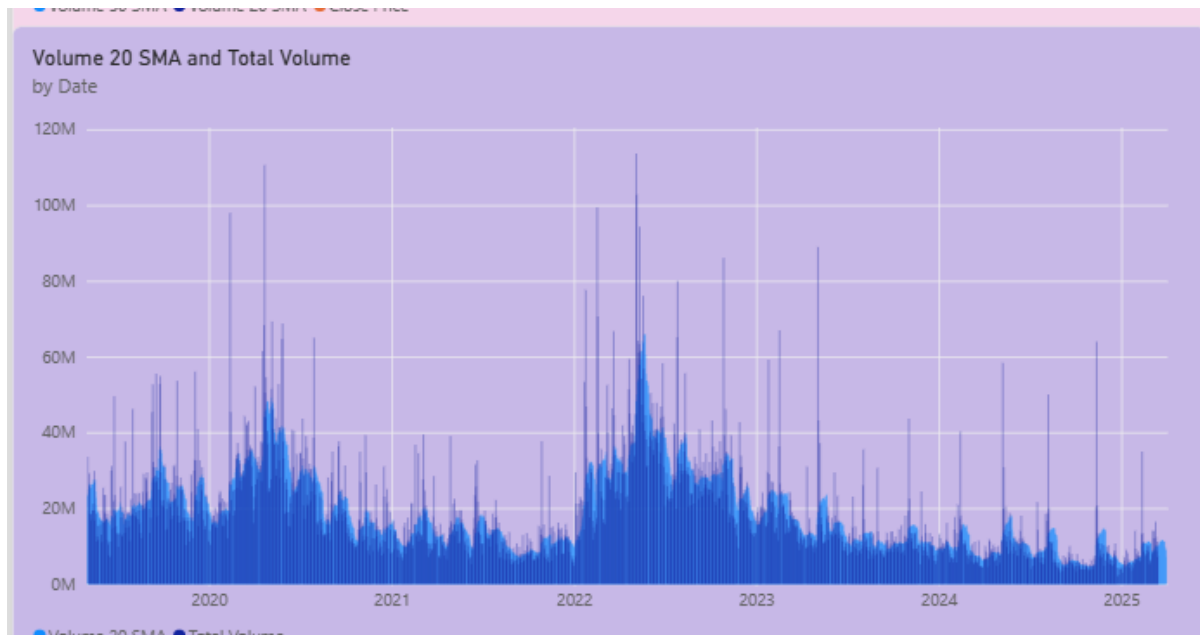
Volume 50 SMA, Volume 20 SMA and Close Price



What this visual shows

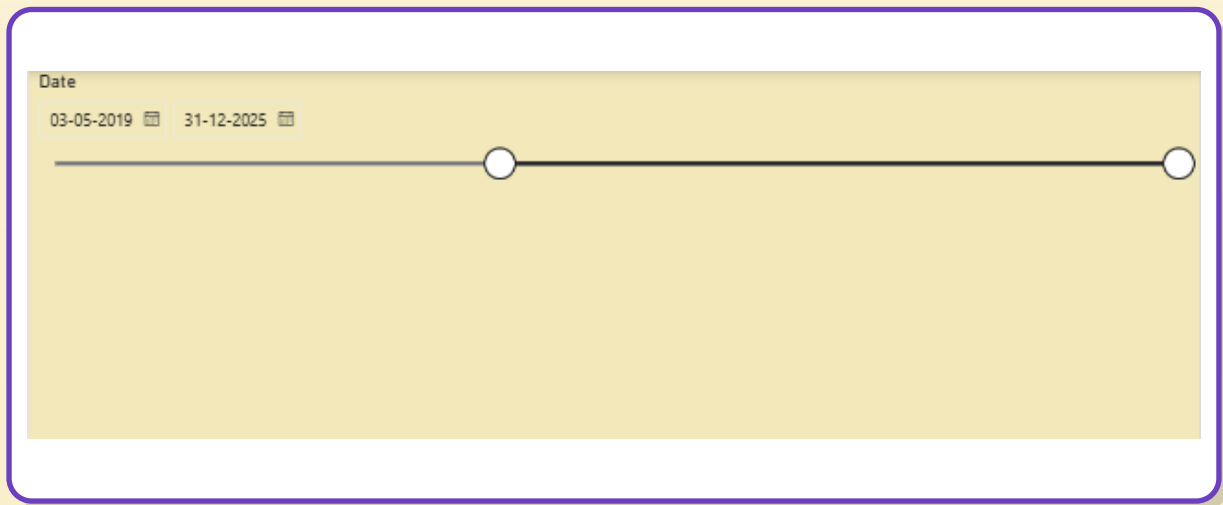
- This is a combo chart (stacked area + line) plotted against the Date axis from 2019 to 2025.
- The dark blue and light blue stacked areas represent the Volume 50-day Simple Moving Average and the Volume 20-day Simple Moving Average, measured on the left axis (in millions).
- The orange line overlaid on top represents the daily Close Price of the stock, measured on the right axis.
- Sharp spikes in the volume areas (e.g. around 2020 and early 2022) indicate periods of unusually heavy trading, which often coincide with major price movements.
- The chart lets viewers visually correlate trading activity (volume) with price direction — rising volume alongside a rising price line suggests strong buying interest.

Volume 20 SMA and Total Volume



What this visual shows

- This is a combination chart showing the Volume 20-day SMA (smoothed trend line/area) together with the raw Total Volume traded on each day, plotted by Date.
- The tall thin spikes represent single-day trading volume, while the smoother shaded band underneath represents the 20-day moving average — filtering out day-to-day noise.
- Recurring volume clusters can be seen around 2020, 2022 and again picking up near 2025, suggesting periodic bursts of market activity.
- This chart complements Visual 1 by giving a clearer, unfiltered view of raw daily volume spikes that get smoothed out in the moving-average chart above.
- Together, both volume visuals help identify accumulation or distribution phases in the stock's trading history.



What this control does

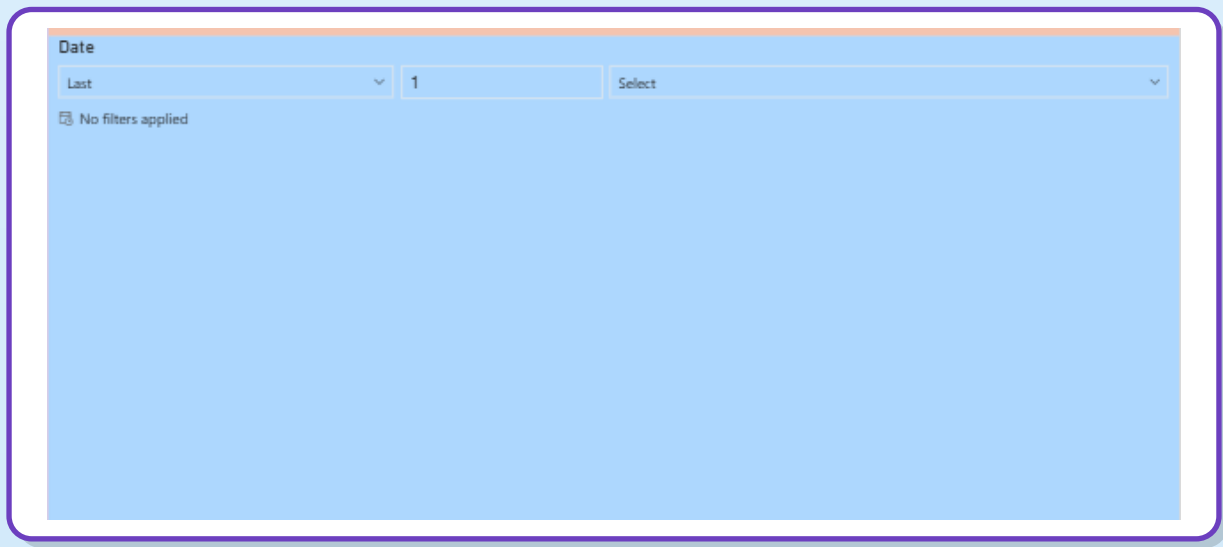
- This is a between-range slider filter bound to the Date field, currently set to span 03-05-2019 to 31-12-2025.
- The two circular handles can be dragged independently to narrow the report to any custom time window, letting the user 'zoom in' on a specific period of interest.
- Every visual on the report page (both charts and the KPI card) responds dynamically to changes made here, since all visuals share the same underlying Date field.
- This makes the dashboard interactive rather than static — the same report can answer 'what happened in 2020?' or 'what happened in the last 6 months?' with a single drag.

Price metric



What this control does

- This is a checkbox-style slicer offering four selectable price metrics: Close Price, High Price, Low Price and Open Price.
- In the screenshot, High Price is checked (filled black box), meaning the report is currently configured to reference the daily High Price wherever a 'price metric' measure is used.
- This slicer allows the user to switch the perspective of the analysis — for example comparing volume against the day's High instead of its Close — without editing the report.
- It is a single-select style filter here (only one metric appears active at a time), keeping the visuals focused on one consistent price measure.



What this control does

- This is a relative date filter card, made up of three parts: a comparison dropdown ('Last'), a numeric input (currently '1'), and a unit dropdown ('Select').
- It lets the user quickly filter the report to a rolling window such as 'Last 1 Year', 'Last 7 Days' or 'Next 3 Months' simply by picking values from the dropdowns.
- The status line 'No filters applied' confirms that this particular relative filter is currently inactive, and the report is showing the full date range instead.
- This control is a faster alternative to the drag-slider on Page 4 for users who want a precise, typed relative period rather than dragging handles.

Close Price, High Price, Open Price, Total Volume, Daily



Close Price	High Price	Open Price	Total Volume	Daily
48.46	176.29	48.46	40912544200	

What this visual shows

- This second screenshot is a compact summary table (card/matrix visual) displaying five key figures for the currently selected date: Close Price, High Price, Open Price, Total Volume and a 'Daily' indicator.
- For the selected day, the values read Close Price = 48.46, High Price = 176.29, Open Price = 48.46, and Total Volume = 40,912,544,200 shares.
- The horizontal scrollbar at the bottom suggests this card is part of a wider table that may contain additional columns (such as Low Price or Date) not visible in this cropped view.
- Because Close and Open Price are identical (48.46) while High Price is much larger (176.29), this row may represent an aggregated or unusual trading day worth further investigation.
- This card acts as a quick-reference snapshot that updates alongside the filters shown on the earlier pages, giving an at-a-glance numeric summary to accompany the visual charts.

Conclusion

The Power BI dashboard examined in this report brings together price and volume data into a single, interactive view of stock market behaviour between 2019 and 2025.

The two main charts show how trading volume (through its 20-day and 50-day moving averages) tends to spike alongside major price movements, helping to visually connect market activity with price trends over time. The raw Total Volume chart adds further detail by revealing individual high-volume trading days that the smoothed averages alone would otherwise hide.

The filter panels — the date range slider, the relative date filter, and the price metric selector — turn the dashboard from a static picture into a flexible analytical tool. A user can narrow the timeframe, switch between Open/High/Low/Close price metrics, or jump to a relative period such as the last year, and every visual updates instantly to match.

Finally, the key metrics summary card ties everything together by surfacing the exact numeric values (Close, High, Open Price and Total Volume) behind whatever date is currently selected, giving both a visual and a numeric perspective on the same underlying data.

Overall, this dashboard demonstrates good practice in interactive report design: it pairs clear visual storytelling (trend and volume charts) with precise, user-controlled filtering, making it a useful tool for anyone analysing stock price and volume behaviour over time.